

Qualification Pack



Smart Farming Supervisor

QP Code: CUTM/AGR/Q1102

Version: 1.0

NSQF Level: 5

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CUTM/AGR/Q1102: Smart Farming Supervisor

Brief Job Description

The individual manages the crop in the farm using smart tools and latest technology. He or she is responsible to understand field variability using IoT based sensors and UAV. Further, he should devise appropriate strategies to manage the identified variability precisely, synchronizing to the crop demand.

Personal Attributes

A Smart Farming Supervisor should be innovative, tech-savvy, and adaptable, with strong problem-solving and decision-making skills. They must have the ability to operate and monitor modern agricultural technologies such as IoT devices, drones, sensors, and automated equipment. Analytical thinking and data interpretation skills are essential to optimize crop yield, resource use, and sustainability. Leadership qualities and team management skills are needed to guide farm workers in adopting precision farming methods. Good communication and planning abilities help in coordinating with stakeholders, agri-tech companies, and farmers. Being environmentally conscious, detail-oriented, and committed to digital transformation in agriculture are key attributes.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N1105: Preparation of Land Using Smart Tools and Equipment](#)
2. [CUTM/AGR/N1106: Knowledge of Smart Seed and Sowing](#)
3. [CUTM/AGR/N1107: Understand Smart Nutrient Management](#)
4. [CUTM/AGR/N1108: Study of Smart Irrigation Water and Weed Management](#)
5. [CUTM/AGR/N1109: Monitoring and Management of Insects and Diseases Management using Smart Technology](#)
6. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Advanced Farming

Qualification Pack

Occupation	Farm machinery, Equipment Operation and maintenance
Country	India
NSQF Level	5
Credits	22
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (in Agriculture/ Horticulture) with NA of experience OR Completed 1st year of UG (UG Certificate) (in Agriculture/ Horticulture) with 1.5 years of experience relevant experience OR 12th grade Pass with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-05-AG-04287-2025-V1-CUTM
NQR Version	1.0

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CUTM/AGR/N1105: Preparation of Land Using Smart Tools and Equipment

Description

This NOS focuses on preparing farmland using advanced smart technologies to ensure precision and efficiency. It covers the use of drones with multispectral cameras for field imaging, image processing for soil variability analysis, and selection of suitable tillage equipment for optimum seedbed preparation. Learners are trained in installing and calibrating soil moisture sensors, operating IoT panels, and setting up micro-irrigation systems aligned with crop needs. The NOS emphasizes integrating smart tools for uniform fertilizer application, data storage on cloud platforms, and adopting practices that minimize water stagnation, thereby enhancing productivity and sustainability.

Scope

The scope covers the following :

- This NOS focuses on modern land preparation techniques integrating smart farming technologies such as drones, multispectral cameras, IoT-based soil moisture sensors, and image processing software. It covers assessment of field variability, selection of appropriate tillage methods, creation of optimum seedbeds, installation of micro-irrigation systems, and precise basal fertilizer application. The scope emphasizes sustainable practices, water management, and data-driven decisions to enhance soil health, productivity, and resource efficiency, enabling technical managers to optimize farm operations using advanced digital and mechanized tools.

Elements and Performance Criteria

Preparation of land using Smart Tools and Equipment

To be competent, the user/individual on the job must be able to:

- PC1.** Fly the drone equipped with multi spectral camera and capture images.
- PC2.** Collect the captured images and perform image stitching.
- PC3.** Process those stitched images in an image processing software.
- PC4.** Based on the results obtained from image processing determine the right tillage equipment and number of tillages required to obtain optimum tilth.
- PC5.** Fix the appropriate gradient in the fields to avoid water stagnation.
- PC6.** Divide the site into grids based on its variability for installation of soil moisture sensors.
- PC7.** Calibrate the soil moisture sensors and fix a threshold for scheduling irrigation based on crop.
- PC8.** Understand circuit of IoT panel to which soil moisture sensors were connected.
- PC9.** Fix the micro-irrigation set up along the gradient based on the crop.
- PC10.** Connect the output points in the IoT panel to the motor connected to micro-irrigation set up.
- PC11.** Assure uniform application of recommended fertilizers as basal dose.
- PC12.** Check the sensor data is stored in cloud.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** the role of land preparation in crop productivity.
- KU2.** ploughing methods, types of tillage implements, and achieving the ideal soil tilth for seedbed preparation.
- KU3.** how to operate drones equipped with multispectral cameras.
- KU4.** how to capture aerial images, stitching them, and applying image processing techniques for land assessment.
- KU5.** image processing software and steps involved in analyzing stitched aerial images.
- KU6.** how to interpret results to identify soil variability and suitability for cultivation.
- KU7.** how to use image-derived data to recommend appropriate tillage equipment and the number of tillages required.
- KU8.** how to fix appropriate gradients in the field to prevent water stagnation.
- KU9.** how to divide the farmland into grids for systematic monitoring.
- KU10.** how to install soil moisture sensors at grid points for precise soil-water management.
- KU11.** the circuit and functioning of IoT panels.
- KU12.** how to connect soil moisture sensors to IoT panels and calibrating them to fix irrigation thresholds based on crop requirements.
- KU13.** how to fix micro-irrigation systems along gradients.
- KU14.** how to link irrigation motors and pumps with IoT panels for automated and efficient irrigation scheduling.
- KU15.** cloud platforms (Amazon, Google, etc.) for storing sensor and image data.
- KU16.** how to ensure accessibility and secure data management for real-time decision-making.
- KU17.** the uniform application of recommended basal fertilizers.
- KU18.** how to ensure integration of nutrient application with sensor-based and IoT-monitored systems.
- KU19.** how to follow safety guidelines while handling drones, sensors, and farm equipment.
- KU20.** how to maintain proper calibration, troubleshooting, and ensuring compliance with environmental and workplace standards.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** operate drones with multispectral cameras, IoT panels, soil moisture sensors, and image processing software to assess land conditions and variability effectively.
- GS2.** process aerial images, stitching datasets, and interpreting results to recommend suitable tillage methods, gradient fixing, and land preparation strategies.
- GS3.** divide land into grids, calibrate soil moisture sensors, and set irrigation thresholds based on crop-specific requirements, ensuring accurate and efficient resource use.
- GS4.** collect, analyze, and store field data in cloud-based systems, enabling real-time monitoring, retrieval, and informed decision-making.
- GS5.** set up micro-irrigation systems, connecting IoT outputs to pumps/motors, and ensuring uniform fertilizer application as a basal dose.



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- GS6.** adhere to health, safety, and environmental regulations while handling drones, fertilizers, and irrigation systems, ensuring safe and sustainable operations.
- GS7.** maintain detailed work-related records, reports of soil/land data, calibration logs, and operational results in line with organizational and industry standards.
- GS8.** coordinate with agronomists, technicians, and farmers for synchronized land preparation, ensuring efficient integration of smart tools with conventional practices.
- GS9.** adopt new farming technologies, software upgrades, and updated smart farming practices for enhanced productivity and sustainability.
- GS10.** evaluate field conditions, troubleshooting equipment/software issues, and selecting optimal techniques, with accountability for quality outcomes in land preparation.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparation of land using Smart Tools and Equipment</i>	40	40	-	20
PC1. Fly the drone equipped with multi spectral camera and capture images.	5	4	-	2
PC2. Collect the captured images and perform image stitching.	2	2	-	-
PC3. Process those stitched images in an image processing software.	2	2	-	-
PC4. Based on the results obtained from image processing determine the right tillage equipment and number of tillages required to obtain optimum tilth.	2	5	-	3
PC5. Fix the appropriate gradient in the fields to avoid water stagnation.	5	5	-	1
PC6. Divide the site into grids based on its variability for installation of soil moisture sensors.	4	6	-	1
PC7. Calibrate the soil moisture sensors and fix a threshold for scheduling irrigation based on crop.	5	8	-	2
PC8. Understand circuit of IoT panel to which soil moisture sensors were connected.	4	2	-	3
PC9. Fix the micro-irrigation set up along the gradient based on the crop.	4	2	-	1
PC10. Connect the output points in the IoT panel to the motor connected to micro-irrigation set up.	3	1	-	1
PC11. Assure uniform application of recommended fertilizers as basal dose.	2	1	-	2
PC12. Check the sensor data is stored in cloud.	2	2	-	4
NOS Total	40	40	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1105
NOS Name	Preparation of Land Using Smart Tools and Equipment
Sector	Agriculture
Sub-Sector	
Occupation	Farm machinery, Equipment Operation and maintenance
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1106: Knowledge of Smart Seed and Sowing

Description

This NOS focuses on the application of smart technologies and scientific methods for seed selection, crop planning, and precision sowing practices. It involves identifying suitable high-value crops, understanding seed characteristics, and determining appropriate sowing depth and spacing. The role emphasizes the use of modern tools such as seed drills, IoT-enabled devices, and drones for ensuring uniform sowing, better germination, and optimized plant population. It also integrates knowledge of crop modeling, data-driven decision-making, and sustainable practices to enhance productivity, reduce resource wastage, and improve farm profitability.

Scope

The scope covers the following :

- This NOS covers the competencies required for selecting, testing, and sowing seeds using smart technologies to ensure precision and higher productivity. It includes the ability to install and use decision-support software for crop modelling, identify suitable high-value crops, and procure quality seeds from reliable sources. Learners are trained to test seed purity, assess germination percentage, measure 1000-grain weight, and determine optimum sowing spacing. The NOS also emphasizes calibration of seed drills or robotic planters, and executing sowing at the correct soil depth using mechanized and IoT-enabled tools. The scope ensures that the trainee develops both theoretical knowledge and practical expertise in smart seed selection and sowing techniques, thereby enhancing efficiency, yield potential, and resource optimization in modern farming systems.

Elements and Performance Criteria

Knowledge of Smart Seed and Sowing

To be competent, the user/individual on the job must be able to:

- PC1.** Install decision support software for crop modelling.
- PC2.** Perform crop modeling and select suitable high value crop.
- PC3.** Procure the seed material from a reliable source.
- PC4.** Test the purity and germination percentage.
- PC5.** Measure the 1000 grain weight using seed counter.
- PC6.** Fix the optimum spacing for sowing.
- PC7.** Calibrate the seed drill/ robotic planter.
- PC8.** Sow the seed at optimum soil depth.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** how to install and operate DSS software for crop modelling to select suitable high-value crops based on soil, climate, and market needs.

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- KU2.** how to identify and choosing appropriate high-value crops through scientific modelling and local suitability studies.
- KU3.** Seed Procurement Standards - Awareness of reliable sources for procuring seed material, ensuring quality standards and certification compliance.
- KU4.** testing methods such as purity testing, germination percentage, and 1000-grain weight measurement using seed counters to ensure seed viability and uniformity.
- KU5.** how to determine and fix optimum spacing between seeds to enhance plant growth, reduce competition, and improve yield efficiency.
- KU6.** Calibration of Smart Sowing Equipment - Knowledge of calibration techniques for seed drills and robotic planters to ensure precision in seed placement and uniform distribution.
- KU7.** the importance of correct sowing/planting depth for different crops, ensuring proper germination and root establishment.
- KU8.** Application of Smart Tools in Sowing - Familiarity with precision sowing technologies, robotics, and IoT-enabled planters for efficient resource utilization.
- KU9.** how seed and sowing operations are linked with soil analysis, nutrient management, and irrigation systems to achieve precision farming outcomes.
- KU10.** how smart sowing techniques contribute to sustainable agriculture by optimizing seed use, reducing wastage, and improving crop productivity in a cost-effective manner.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** to install, configure, and operate crop modeling software to analyze climatic, soil, and resource conditions for selecting suitable high-value crops.
- GS2.** evaluate multiple crop options through smart tools, assessing profitability, and recommending the best-suited variety for a given agro-climatic zone.
- GS3.** identify reliable seed sources, testing seed purity, and evaluating germination percentage for ensuring high productivity.
- GS4.** use seed counters and other devices to measure parameters like 1000-grain weight, which is essential for yield prediction and sowing calibration.
- GS5.** the optimum depth and spacing requirements for different crops, ensuring proper plant population and uniform growth.
- GS6.** calibrate and operate smart sowing equipment like seed drills and robotic planters for precision planting.
- GS7.** work with IoT-linked sowing equipment that adjusts sowing parameters (depth, spacing, seed rate) automatically based on soil data.
- GS8.** perform sowing/transplanting operations accurately, aligning manual expertise with smart farming tools for efficiency.
- GS9.** Maintain systematic records of seed quality tests, sowing parameters, and field performance data for decision-making and compliance.
- GS10.** Develop troubleshooting skills to handle sowing equipment malfunctions, variability in seed performance, and adapting sowing methods to real-time field conditions.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Knowledge of Smart Seed and Sowing</i>	40	50	-	10
PC1. Install decision support software for crop modelling.	5	10	-	1
PC2. Perform crop modeling and select suitable high value crop.	5	5	-	2
PC3. Procure the seed material from a reliable source.	5	5	-	1
PC4. Test the purity and germination percentage.	5	5	-	1
PC5. Measure the 1000 grain weight using seed counter.	5	5	-	1
PC6. Fix the optimum spacing for sowing.	5	5	-	1
PC7. Calibrate the seed drill/ robotic planter.	5	5	-	1
PC8. Sow the seed at optimum soil depth.	5	10	-	2
NOS Total	40	50	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1106
NOS Name	Knowledge of Smart Seed and Sowing
Sector	Agriculture
Sub-Sector	
Occupation	Farm machinery, Equipment Operation and maintenance
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1107: Understand Smart Nutrient Management

Description

This NOS focuses on the scientific and technology-driven management of soil nutrients to optimize crop productivity. It involves the use of nutrient-specific sensors and IoT-based systems to monitor soil fertility in real-time. The learner is expected to generate and analyze soil fertility maps using GIS tools, calibrate Variable Rate Technology (VRT), and devise nutrient application maps tailored to crop needs. The role also includes integrating data into decision support systems to determine the right source, dose, and timing of nutrient application, ensuring precision, sustainability, and cost-effectiveness. This NOS empowers the trainee to adopt smart nutrient management practices that enhance soil health, reduce wastage of inputs, and improve overall farm efficiency.

Scope

The scope covers the following :

- It covers the operation of nutrient-specific sensors, integration with IoT platforms, and development of soil fertility maps through GIS tools. Learners are trained to calibrate Variable Rate Technology (VRT) equipment and prepare nutrient application maps for site-specific fertilizer use. The scope also includes data collection for decision support systems (DSS), using DSS to determine the right source, type, dose, and timing of nutrients, and ensuring efficient and sustainable nutrient management practices.
- By applying precision farming tools, this NOS ensures improved crop yield, soil health maintenance, and reduced input wastage, aligning with environmentally sustainable and economically viable smart farming practices.

Elements and Performance Criteria

Understand Smart Nutrient Management

To be competent, the user/individual on the job must be able to:

- PC1.** Install nutrient specific sensors.
- PC2.** Connect the sensor data to IoT.
- PC3.** Develop soil fertility maps using GIS software.
- PC4.** Calibrate the Variable Rate Technology (VRT).
- PC5.** Devise nutrient application map and feed it to VRT.
- PC6.** Collect the data to train the decision support system.
- PC7.** Use Decision Support Systems to identify the right source and dose of application of nutrients.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** macro and micro-nutrients, their role in plant growth, and crop-specific nutrient needs.
- KU2.** the use of nutrient-specific sensors - knowledge of advanced tools and IoT-enabled devices to measure soil nutrient levels accurately.

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- KU3.** Soil fertility mapping - ability to generate and interpret soil fertility maps using GIS software and digital platforms.
- KU4.** Smart nutrient application methods - awareness of precision farming techniques for site-specific and crop-specific nutrient delivery.
- KU5.** Integration with IoT and cloud-based platforms - understanding how to access, analyze, and utilize sensor data for nutrient decision-making.
- KU6.** Fertilizer scheduling and optimization - knowledge of appropriate timing, dosage, and method of nutrient application to enhance efficiency and reduce wastage.
- KU7.** Sustainable nutrient management practices - understanding of balanced fertilization, organic nutrient integration, and environmental safety measures.
- KU8.** Image processing and remote sensing in nutrient analysis - awareness of digital tools for assessing crop nutrient deficiencies through aerial imagery.
- KU9.** Data interpretation and decision support systems (DSS) - skills to analyze sensor-based and GIS-based data for practical nutrient management decisions.
- KU10.** Compliance with smart farming standards - knowledge of modern agricultural practices, eco-friendly nutrient use, and adherence to safety norms.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** install, operate, and calibrate nutrient-specific sensors and connect them with IoT systems for real-time monitoring and data collection.
- GS2.** use GIS software to develop detailed soil fertility maps for precision nutrient planning.
- GS3.** calibrate and operate VRT devices for site-specific nutrient application, reducing wastage and improving efficiency.
- GS4.** collect, analyze, and interpret field and sensor data to train Decision Support Systems (DSS) for nutrient management.
- GS5.** devise nutrient application maps and integrate them with precision agriculture tools for optimal crop growth.
- GS6.** select the right nutrient sources, determining the appropriate dosage, and scheduling applications based on crop requirements.
- GS7.** do eco-friendly nutrient application practices that enhance productivity while minimizing environmental impact.
- GS8.** coordinate with farm staff, share nutrient management plans clearly, and document outcomes for evaluation and compliance.
- GS9.** troubleshoot issues related to sensor failures, data inconsistencies, or unexpected soil nutrient variations and adapt strategies accordingly.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand Smart Nutrient Management</i>	40	50	-	10
PC1. Install nutrient specific sensors.	7	9	-	2
PC2. Connect the sensor data to IoT.	7	9	-	1
PC3. Develop soil fertility maps using GIS software.	8	9	-	1
PC4. Calibrate the Variable Rate Technology (VRT).	8	8	-	1
PC5. Devise nutrient application map and feed it to VRT.	4	8	-	1
PC6. Collect the data to train the decision support system.	3	4	-	2
PC7. Use Decision Support Systems to identify the right source and dose of application of nutrients.	3	3	-	2
NOS Total	40	50	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1107
NOS Name	Understand Smart Nutrient Management
Sector	Agriculture
Sub-Sector	
Occupation	Farm machinery, Equipment Operation and maintenance
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1108: Study of Smart Irrigation Water and Weed Management

Description

This NOS focuses on the application of smart technologies for efficient water use and weed control in modern farming systems. It involves the installation and calibration of IoT-based soil moisture sensors, adoption of suitable micro-irrigation methods, and programming irrigation schedules using variable rate technology to optimize water distribution as per crop needs. Learners are trained to differentiate weeds from crops using image processing software, spectral analysis, and remote sensing tools, enabling targeted weed management. The NOS also emphasizes the judicious use of herbicides at the right time and dosage, ensuring sustainable crop production. Through this NOS, trainees gain competence in precision irrigation, water conservation, and integrated weed control measures, thereby enhancing productivity while reducing input costs and environmental impacts.

Scope

The scope covers the following :

- It covers the installation and calibration of IoT-based soil moisture sensors, scheduling irrigation through micro-irrigation and variable rate technologies, and integrating data-driven systems for precise water use. Learners will also gain expertise in distinguishing weeds from crops using image processing software, spectral wavelength differences, and remote sensing tools, followed by the application of suitable herbicides at the correct stage. The scope extends to enhancing water-use efficiency, reducing crop stress, minimizing weed competition, and optimizing input costs, thereby ensuring sustainable agricultural practices. This NOS is critical in promoting resource conservation, higher productivity, and environmentally safe weed management strategies, aligning with modern precision farming systems.

Elements and Performance Criteria

Study of Smart Irrigation Water and Weed Management

To be competent, the user/individual on the job must be able to:

- PC1.** Install IoT based soil moisture sensors across the farm.
- PC2.** Adopt the right method of micro-irrigation.
- PC3.** Code the IoT based sensors for threshold and duration of irrigation schedule.
- PC4.** Calibrate the system before installation.
- PC5.** Install the calibrated irrigation system in the field.
- PC6.** Schedule the application using variable rate technology.
- PC7.** Distinguish weed species from crop by their differences in wavelength.
- PC8.** Use image processing soft wares to differentiate weeds from crop.
- PC9.** Ascertain the type of weeds that will affect the crop.
- PC10.** Use appropriate herbicides in the appropriate time.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** IoT-based Soil Moisture Monitoring - Knowledge of installing, coding, and calibrating IoT soil moisture sensors to assess real-time soil water status and optimize irrigation schedules.
- KU2.** Smart Irrigation Systems - Understanding of micro-irrigation techniques (drip, sprinkler, etc.), their design, installation, and operation for efficient water use in different crops.
- KU3.** Irrigation Scheduling with Technology - Ability to use Variable Rate Technology (VRT) and sensor data to schedule irrigation precisely as per crop requirement and soil condition.
- KU4.** Calibration and Maintenance - Knowledge of calibrating irrigation systems before installation and ensuring uniform water distribution across the field.
- KU5.** Crop Water Requirement - Understanding crop-specific water demand and how different environmental and soil factors affect irrigation needs.
- KU6.** Weed Identification Techniques - Ability to distinguish weeds from crops using multispectral imaging, wavelength differences, and image processing software.
- KU7.** Types and Characteristics of Weeds - Familiarity with common weed species, their growth behavior, and their impact on crop yield.
- KU8.** Weed Management Approaches - Knowledge of integrated weed management practices including manual, mechanical, chemical, and smart sensor-based methods.
- KU9.** Herbicide Use and Safety - Understanding the types of herbicides, correct dosage, application timing, and safety measures to minimize crop damage and environmental hazards.
- KU10.** Decision-making through Smart Tools - Ability to interpret IoT and imaging data for making informed decisions on water conservation and weed control strategies, ensuring sustainability and cost-effectiveness.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** to install, calibrate, and manage IoT-based soil moisture sensors and micro-irrigation systems for precise water scheduling.
- GS2.** use image processing software and spectral analysis to distinguish crops from weeds for accurate weed management.
- GS3.** do water resource management - Understanding of crop-specific water requirements and the skill to plan irrigation schedules using smart technologies and variable rate applications.
- GS4.** identify weed species, evaluate their impact, and apply herbicides or eco-friendly weed management practices at the right time and dose.
- GS5.** analyze soil moisture and crop data to make timely decisions on irrigation, detect anomalies, and adopt corrective measures.
- GS6.** Prepare field records, irrigation schedules, and weed management reports following industry standards and digital formats.
- GS7.** Ensure safe handling of herbicides, use of personal protective equipment (PPE), and adoption of environmentally sustainable farming practices.
- GS8.** communicate with farm teams, suppliers, and supervisors effective teamwork for coordinated water and weed management.



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- GS9.** Use IoT dashboards, GIS-based tools, and cloud platforms for real-time monitoring, analysis, and decision support.
- GS10.** Stay updated with advancements in smart farming technologies, irrigation automation, and integrated weed management strategies.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Study of Smart Irrigation Water and Weed Management</i>	30	60	-	10
PC1. Install IoT based soil moisture sensors across the farm.	3	6	-	1
PC2. Adopt the right method of micro-irrigation.	3	6	-	1
PC3. Code the IoT based sensors for threshold and duration of irrigation schedule.	3	6	-	1
PC4. Calibrate the system before installation.	3	6	-	1
PC5. Install the calibrated irrigation system in the field.	3	6	-	1
PC6. Schedule the application using variable rate technology.	3	6	-	1
PC7. Distinguish weed species from crop by their differences in wavelength.	3	6	-	1
PC8. Use image processing soft wares to differentiate weeds from crop.	3	6	-	1
PC9. Ascertain the type of weeds that will affect the crop.	3	6	-	1
PC10. Use appropriate herbicides in the appropriate time.	3	6	-	1
NOS Total	30	60	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1108
NOS Name	Study of Smart Irrigation Water and Weed Management
Sector	Agriculture
Sub-Sector	
Occupation	Farm machinery, Equipment Operation and maintenance
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1109: Monitoring and Management of Insects and Diseases Management using Smart Technology

Description

This NOS covers the application of smart technologies for the timely detection, monitoring, and management of pests and diseases in crops. It involves the use of drones, multispectral scanners, and image processing software to remotely sense and analyze farm conditions, enabling early identification of insect infestations and disease symptoms. The individual is required to document pest behavior, prepare pest calendars, and determine the stage and part of the crop vulnerable to attacks. Integrated pest and disease management techniques, including UAV-based spraying, resistant varieties, and preventive measures such as pheromone and light traps, are emphasized to minimize crop loss. The NOS also ensures judicious and safe use of agrochemicals in right quantities, promoting precision agriculture and sustainable farming practices.

Scope

The scope covers the following :

- This NOS covers the application of smart technologies such as drones, multispectral scanners, IoT-based sensors, and image processing tools for the early detection, monitoring, and management of insect pests and crop diseases. It emphasizes accurate identification of pest and disease incidence, assessment of crop growth stages, and the adoption of preventive and curative strategies. The scope also includes preparing pest calendars, implementing Integrated Pest Management (IPM) practices, and using UAV-based interventions and resistant crop varieties to ensure sustainable and precise crop protection.

Elements and Performance Criteria

Monitoring and Management of Insects and Diseases Management using Smart Technology

To be competent, the user/individual on the job must be able to:

- PC1.** Remote sense the farm using multispectral scanners.
- PC2.** Analyze the images in an image processing software.
- PC3.** Identify the pest.
- PC4.** Find out the level of incidence.
- PC5.** Identify the stage of crop growth when the pests attack.
- PC6.** Identify the part of the crop where the specific pests will attack.
- PC7.** Identify and document the pest behavior.
- PC8.** Prepare a pest calendar with all pest characteristics.
- PC9.** Document the preventive measures for the pests.
- PC10.** Adopt UAV based pest control when the incidence level is above ETL.
- PC11.** Identify the stage of crop when it is vulnerable to disease.
- PC12.** Identify the symptoms for the diseases.
- PC13.** Identify the mode of disease transfer.

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- PC14.** Ascertain the appropriate integrated pest management techniques.
- PC15.** Use appropriate pest and disease management chemicals in right quantities.
- PC16.** Use pest and disease resistant varieties.
- PC17.** Practice preventive techniques like pheromone traps, light traps, bird perches, sticky traps, etc. according to the cultivar.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Remote Sensing for Pest/Disease Monitoring - Understanding the use of drones and multispectral scanners to remotely sense crop health and identify stress symptoms caused by pests and diseases.
- KU2.** Image Processing and Data Interpretation - Knowledge of image stitching, processing software, and interpretation techniques to detect pest infestations and disease occurrence from aerial and ground-based data.
- KU3.** Pest Identification & Incidence Analysis - Familiarity with major crop pests, their life cycles, incidence levels, stages of crop vulnerability, and affected plant parts for precise monitoring.
- KU4.** Disease Recognition - Understanding the symptoms of major crop diseases, their stages of occurrence, and different modes of disease transmission (air, soil, water, vectors).
- KU5.** Pest & Disease Calendars - Ability to document pest and disease behavior, prepare calendars for monitoring seasonal outbreaks, and integrate them into farm management systems.
- KU6.** Preventive Measures & Trapping Techniques - Knowledge of eco-friendly preventive practices like pheromone traps, light traps, sticky traps, bird perches, and their crop-specific applications.
- KU7.** Integrated Pest & Disease Management (IPDM) - Awareness of integrated approaches including cultural, biological, and chemical methods, and how to adopt them using smart farming tools.
- KU8.** UAV-based Control Measures - Understanding how to apply pesticides, fungicides, and biocontrol agents using drones (UAVs) when infestation crosses the Economic Threshold Level (ETL).
- KU9.** Use of Resistant Varieties & Agrochemicals - Familiarity with resistant crop varieties, and safe, precise use of agrochemicals (pesticides, fungicides, insecticides) in correct doses and timings.
- KU10.** Smart Decision-Support Systems - Ability to integrate IoT, GIS, and DSS (Decision Support Systems) for predictive pest/disease alerts, mapping, and data-driven management strategies.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use UAVs, multispectral scanners, IoT-enabled devices, and image-processing software for real-time monitoring of crop health and pest incidence.
- GS2.** analyze remote-sensed images, identifying pests/diseases, assessing incidence levels, and documenting symptoms for informed decision-making.

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- GS3.** implement applications of Integrated Pest Management (IPM) techniques, including resistant varieties, pheromone traps, light traps, sticky traps, and biological control methods.
- GS4.** develop pest calendars, document preventive strategies, and apply chemical/organic solutions in the right dosage and timing.
- GS5.** evaluate incidence thresholds (ETL) and decide when to adopt UAV-based pest control interventions to prevent economic losses.
- GS6.** Prepare accurate records on pest behavior, preventive measures, and disease transmission, ensuring compliance with agricultural standards and traceability.
- GS7.** Work effectively with farmers, agronomists, and technical teams, communicating findings and solutions clearly for timely action.
- GS8.** Apply eco-friendly pest and disease management practices that reduce chemical dependency and promote long-term soil and crop health.
- GS9.** Troubleshoot field-level challenges such as sensor malfunctions, unexpected pest outbreaks, or crop vulnerabilities through adaptive strategies.
- GS10.** Ensure safe handling of chemicals, UAV operations, and digital tools while following organizational health, safety, and environmental guidelines.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Monitoring and Management of Insects and Diseases Management using Smart Technology</i>	40	40	-	20
PC1. Remote sense the farm using multispectral scanners.	3	3	-	1
PC2. Analyze the images in an image processing software.	2	1	-	1
PC3. Identify the pest.	2	2	-	1
PC4. Find out the level of incidence.	2	2	-	2
PC5. Identify the stage of crop growth when the pests attack.	3	3	-	1
PC6. Identify the part of the crop where the specific pests will attack.	3	3	-	1
PC7. Identify and document the pest behavior.	3	3	-	1
PC8. Prepare a pest calendar with all pest characteristics.	3	3	-	1
PC9. Document the preventive measures for the pests.	2	3	-	1
PC10. Adopt UAV based pest control when the incidence level is above ETL.	2	3	-	1
PC11. Identify the stage of crop when it is vulnerable to disease.	2	1	-	1
PC12. Identify the symptoms for the diseases.	2	2	-	1
PC13. Identify the mode of disease transfer.	1	3	-	1
PC14. Ascertain the appropriate integrated pest management techniques.	1	3	-	1
PC15. Use appropriate pest and disease management chemicals in right quantities.	3	3	-	2
PC16. Use pest and disease resistant varieties.	3	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. Practice preventive techniques like pheromone traps, light traps, bird perches, sticky traps, etc. according to the cultivar.	3	1	-	2
NOS Total	40	40	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1109
NOS Name	Monitoring and Management of Insects and Diseases Management using Smart Technology
Sector	Agriculture
Sub-Sector	
Occupation	Farm machinery, Equipment Operation and maintenance
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 60% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

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Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N1105.Preparation of Land Using Smart Tools and Equipment	40	40	0	20	100	15
CUTM/AGR/N1106.Knowledge of Smart Seed and Sowing	40	50	0	10	100	20
CUTM/AGR/N1107.Understand Smart Nutrient Management	40	50	0	10	100	10
CUTM/AGR/N1108.Study of Smart Irrigation Water and Weed Management	30	60	0	10	100	20
CUTM/AGR/N1109.Monitoring and Management of Insects and Diseases Management using Smart Technology	40	40	0	20	100	15
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	-	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	240	340	-	70	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.